

Multithreading

108. Consider the following code:

```
class TestClass {
    ....public void f0() {
        .....synchronized (this) {
            .....for (int i = 0; i < 5; i++)
            .....System.out.println(Thread.currentThread().getName() + " " + i);
            .....}
        ....}
    ....public void f1() {
        .....synchronized (TestClass.class) {
            .....for (int i = 0; i < 5; i++) {
            .....System.out.println(Thread.currentThread().getName() + " " + i);
            .....}
        .....}
        ....}
    ....public synchronized static void f2() {
        .....for (int i = 0; i < 5; i++) {
        .....System.out.println(Thread.currentThread().getName() + " " + i);
        .....}
        ....}
    ....public synchronized static void f3() {
        .....for (int i = 0; i < 5; i++) {
        .....System.out.println(Thread.currentThread().getName() + " " + i);
        .....}
        ....}
    }
}
class SynchronizedTest {
    ....public static void main(String[] args) {
        .....new Thread(new TestClass().f1, "T1").start();
        .....new Thread(TestClass.f2, "T2").start();
        .....new Thread(TestClass.f3, "T3").start();
        ....}
    }
}
```

- Determine and then write down all the possible outputs of the above code segment, assuming that thread T1 will start executing first.
- If we create two different threads executing methods `f0` and `f1`, will they run in parallel? Why or why not? [\[23-24 5b\]](#) [\[Mark: 10 \]](#) [\[Tag: Java \]](#)

109. What is the problem with the following code segment? Write 2 (two) different ways to fix it. [\[23-24 7a\]](#) [\[Mark: 10 \]](#) [\[Tag: Java \]](#)

```
class MyThread implements Runnable {
    ....List<String> list;
    ....Thread t;
    ....public MyThread(List<String> list) {
        .....this.list = list;
        .....t = new Thread(this);
        .....t.start();
        ....}
    ....public void run() {
        .....for (int i = 1; i <= 10000; i++) {
        .....list.add(String.valueOf(i));
        .....}
    }
}
```

```

.....}
....}
}
public class TestSyncArrayList {
....public static void main(String[] args) throws Exception {
.....List<String> list = new ArrayList<>();
.....MyThread myThread1 = new MyThread(list);
.....MyThread myThread2 = new MyThread(list);
.....myThread1.t.join();
.....myThread2.t.join();
.....System.out.println(list.size());
....}
}

```

110. Complete the following Java code that finds the summation of a 10000-element integer array using an array of 10 threads in parallel. You cannot write any more classes, but you can add variables, constructors, and other methods.
[23-24 7c] [Mark: 15] [Tag: Java]

```

class ParallelSum implements Runnable {
....// your code here
}
public class Main {
....public static void main(String[] args) {
.....java.util.Random random = new java.util.Random();
.....int[] numbers = new int[10000];
.....for (int i = 0; i < numbers.length; i++) {
.....numbers[i] = random.nextInt(10000);
.....}
.....ParallelSum[] parallelSum = new ParallelSum[10];
.....// your code here
....}
}

```

111. You are given the following class:

```

class ThreadTest {
....public void f0() {
.....for (int i = 1; i < 500; i++) {
.....System.out.println(Thread.currentThread().getName() + " " + i);
.....}
....}
....public synchronized void f1() {
.....for (int i = 1; i < 1000; i++) {
.....System.out.println(Thread.currentThread().getName() + " " + i);
.....}
....}
....public synchronized void f2() {
.....for (int i = 1; i < 1500; i++) {
.....System.out.println(Thread.currentThread().getName() + " " + i);
.....}
....}
....public synchronized static void fs1() {
.....for (int i = 1; i < 2000; i++) {
.....System.out.println(Thread.currentThread().getName() + " " + i);
.....}
....}
}

```

```

....public synchronized static void fs2() {
.....for (int i = 1; i < 2500; i++) {
.....System.out.println(Thread.currentThread().getName() + " " + i);
.....}
....}
}

```

You have the following restrictions:

R1: You are not allowed to change ThreadTest class

R2: You can create only one object of ThreadTest class

Now, you have to create the following threads from the main method:

T1: This thread will execute the f1 method

T2: This thread will execute the execute f2 method

T3: This thread will execute the execute f0 method

T4: This thread will execute the execute fs1 method

T5: This thread will execute the execute fs2 method

Design the Java code for the main method. Now, compose the answer to the following questions:

- Which of the threads will run in parallel and why?
- Which of the threads will not run in parallel and why? Can you make them parallel with the other threads by relaxing restriction R2? If yes, what changes do you have to make in your code? [\[22-23 5c\]](#) [Mark: 15] [Tag: Java]

112. You are given a task to find the total count of prime numbers in a randomly generated array of 10000 integers with values between 1 and 5000. You are asked to divide the work equally among five different threads. Write complete Java code to compute the total count of prime numbers by dividing the work equally among five threads. The main thread will wait for the five threads to finish and only print the final count. You can use the following code segment to generate a random number between 1 and 5000. [\[22-23 7c\]](#) [Mark: 15] [Tag: Java]

```

java.util.Random random = new Random();
int x = random.nextInt(5000)+1;

```

113. Describe two different ways of creating threads in Java with short examples. Which one is better and why? Explain the difference between a synchronized method and a synchronized statement. [\[21-22 6a\]](#) [Mark: 10] [Tag: Java, Theory]
114. The following Java code (incomplete) is required to find the minimum of a 1000 integers array using an array of 5 threads in parallel.

```

class ParallelMin implements Runnable {
....// your code
}
class Main {
....public static void main(String[] args) {
.....Random random = new Random();
.....int[] numbers = new int[1000];
.....for (int i = 0; i < numbers.length; i++) {
.....numbers[i] = random.nextInt();
.....}
.....ParallelMin [] parallelMin = new ParallelMin[5];
.....// your code
....}
}

```

You need to construct the complete Java code. You cannot write any more classes, but you can add variables, constructors, and other methods. [21-22 6c] [Mark: 15] [Tag: Java]

115. With Java threads, it is very easy to parallelize computations. Suppose you are in a job interview with Oracle and the interviewer asks you to write Java code to find out summation of 1 to 10000. You can't use any mathematical equation; you can only use loops. But you are asked to divide the work equally among 5 different threads. Write complete Java code to compute the summation of 1 to 10000 by dividing the work equally among 5 different threads. The main thread will wait for the 5 threads to finish and will only print the final summation. [20-21 7c] [Mark: 10] [Tag: Java]

116. Consider the following SharedCounter class:

```
class SharedCounter {
    ...private int counter;
    ...SharedCounter () {
        .....this.counter = 0;
        ....}
    ...private void increment () {
        .....this.counter++;
        ....}
    ...public int get() {
        .....return this.counter;
        ....}
    ...public void count () {
        .....for (int i = 1; i <= 1000; i++) {
            .....increment();
            .....}
        ....}
}
```

Explain with code how you can fix the SharedCounter class so that multiple threads can correctly operate on it. You don't need to write the whole class. [20-21 7d] [Mark: 5] [Tag: Java]

117. You are given the following class:

```
class TestClass {
    ...public void f1() {
        .....for (int i = 0; i < 5; i++) {
            .....System.out.println(i);
            .....try {
                .....Thread.sleep(1000);
            .....} catch (InterruptedException e) {
                .....System.out.println(e);
            .....}
        .....}
    ....}
    ...public static void f2() {
        .....for (int i = 0; i < 5; i++) {
            .....System.out.println(i);
            .....try {
                .....Thread.sleep(2000);
            .....} catch (InterruptedException e) {
                .....System.out.println(e);
            .....}
        .....}
    ....}
}
```

You have to create two threads from the main method. The first thread will execute the `f1` method, and the second thread will execute the `f2` method. Write Java code for the main method. You are not allowed to change the `TestClass`; that means `TestClass` can't extend/implement any class/interface. [\[Jan 21 3a\]](#) [\[Mark: 10 \]](#) [\[Tag: Java \]](#)

118. Mr. P and Mr. C worked in the same warehouse. Their work is to put and get products in a shared inventory of size 4. Mr. P puts a single product in the inventory, and Mr. C gets a single product from the inventory at a time. Mr. P cannot put a product if the inventory is full with 4 products. Mr. C cannot get a product if the inventory contains no product. Mr. P can put a product in the inventory if any one of the inventory places is empty. Mr. C can get a product from the inventory if any one of the inventory places is full with a product. Write complete Java code to solve the problem mentioned above using the concepts of inter-thread communication. You have to use `wait` and `notify`/`notifyAll`, and you are not allowed to use `BlockingQueue`. [\[Jan 21 3b\]](#) [\[Mark: 20 \]](#) [\[Tag: Java \]](#)

119. What are two common ways to achieve synchronization between Threads? Which one gives you more control over the synchronization? [\[Jan 21 3c\]](#) [\[Mark: 5 \]](#) [\[Tag: Java, Theory \]](#)

120. With Java threads, it is very easy to parallelize computations. Suppose you are in a job interview and the interviewer asks you to write Java code to find out summation of 1 to 10000000. You can't use any simple equation; you can only use loops. But you are asked to divide the work equally among 10 different threads. Write complete Java code to compute the summation of 1 to 10000000 by dividing the work equally among 10 different threads. The main thread will wait for the 10 threads to finish and will only print the final summation. [\[Jan 20 7a\]](#) [\[Mark: 10 \]](#) [\[Tag: Java \]](#)

121. Consider two entities, the Writer and the Reader, who share a common buffer that is a single character. The Writer's job is to generate English letters circularly from A to Z. The Reader's job is to read and display them in the output. The Writer can't write new letter if the Reader does not read the already written letter. The Reader can't read if the Writer does not write any new letter. Write a java code to solve the above-mentioned problem using the concept of inter thread communication. You can use `wait`/`notifyAll` or `ArrayBlockingQueue`. [\[Jan 20 7b\]](#) [\[Mark: 10 \]](#) [\[Tag: Java \]](#)

122. Write three different ways to create Threads in Java with short code examples. What is the difference between synchronized method and synchronized statement? [\[Jan 20 7c\]](#) [\[Mark: 10 \]](#) [\[Tag: Java, Theory \]](#)

123. Briefly describe the methods that Java uses for interthread communication. [\[17-18 2c\]](#) [\[Mark: 9 \]](#) [\[Tag: Java, Theory \]](#)

124. What are the advantages of multi-threading over process based multitasking? [\[17-18 2d\]](#) [\[Mark: 6 \]](#) [\[Tag: Java, Theory \]](#)

Generics & Collections

125. Consider the following code segment and its output:

```
public class MyMaxTest {
    ....public static void main(String[] args) {
        .....Long[] n1 = {10L, 20L, 30L, 40L, 50L};
        .....MyMax<Long> mx1 = new MyMax<>(n1);
        .....System.out.println(mx1.max());
        .....Float[] n2 = {50.0f, 20.0f, 40.0f, 10.0f, 30.0f};
        .....MyMax<Float> mx2 = new MyMax<>(n2);
        .....System.out.println(mx2.max());
        .....Double[] n3 = {50.0, 20.0, 40.0, 10.0, 30.0};
        .....MyMax<Double> mx3 = new MyMax<>(n3);
        .....System.out.println(mx3.max());
        .....System.out.println(mx1.sameMax1(mx2));
        .....System.out.println(mx2.sameMax2(mx3));
        ....}
    }
```

Output:

```
50.0
50.0
```

Name	Salary (Tk.)
Sakib Al Hasan	3,00,000
Mashrafe Bin Mortaza	null
Mushfiquir Rahim	2,50,000

Now write Java code for the following:

- Create an appropriate Hash-based collection to store the above information and then store the above information.
- Increase Sakib Al Hasan's salary by 50,000 Tk. [\[17-18 4b\]](#) [\[Mark: 16 \]](#) [\[Tag: Java \]](#)

Server Socket

142. A Fibonacci server can accept multiple Fibonacci clients. The client program will send two integers to the server. The server will calculate the number of Fibonacci numbers between the two integers and send it to the client. The client will show the number as output. The server and client will only communicate through objects. Write a complete Java client-server program to handle the above scenario. [\[22-23 8c\]](#) [\[Mark: 15 \]](#) [\[Tag: Java \]](#)
143. A server program can accept multiple clients. The client program will send two integers to the server. The server will calculate the number of prime numbers between the two integers and send it to the client. The client will show the number as output. Develop a complete Java server-client program to handle the above scenario. [\[21-22 8c\]](#) [\[Mark: 15 \]](#) [\[Tag: Java \]](#)
144. Suppose a client wants to connect to a server on IP address 192.168.1.1 and port 44444. Just draw the scenario of operations needed for the successful connection establishment and data transfer between the server and client. You do not need to write any code. What are the names of the classes generally used to read and write objects in networking? What restriction do they impose on the class? [\[20-21 8c\]](#) [\[Mark: 10 \]](#) [\[Tag: Java, Theory \]](#)
145. Briefly describe the types of TCP Sockets. Write a Java Program that opens a connection to a "whois" port (port 43) on the InterNIC server `whois.internic.net`, sends the command-line argument (a website address) over the socket and then prints the data that is returned. [\[17-18 3a\]](#) [\[Mark: 19 \]](#) [\[Tag: Java \]](#)

Type Wrapper

146. Describe boxing and unboxing in Java. Give examples with primitive `int` and `Integer`. Explain the difference between boxing/unboxing and autoboxing/unboxing with code. When you shouldn't use autoboxing/unboxing? [\[21-22 8a\]](#) [\[Mark: 10 \]](#) [\[Tag: Java, Theory \]](#)
147. Consider the following code segment:

```
public class AutoBoxingUnboxingDemo {
    ....static int f (Integer v) {
        .....return v;
    ....}
    ....public static void main (String args[]) {
        .....Integer a = f(100);
        .....Integer iOb, iOb2;
        .....int i;
        .....iOb = 100;
        .....++iOb;
        .....iOb2 = iOb + (iOb / 3);
        .....i = iOb + (iOb / 3);
        .....Integer intOb = 100;
        .....Double doubleOb = 98.6;
        .....doubleOb = doubleOb + intOb;
```